Lab 6
Creating and Combining Sinusoids in MATLAB

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I. Introduction

This lab had two main objectives. The first was to generate discrete-time (DT) sinusoids in MATLAB and see how three defining parameters (amplitude, frequency, and phase) affected both the appearance of the waveform and the sound it produced, including what happens when a signal is sampled too slowly to represent its frequency. The second was to combine multiple sinusoids at different frequencies, amplitudes, and phases to see how more complex waveforms, such as a square wave, can be formed from a sum of simple sine waves, the idea behind Fourier synthesis.

II. Results

1. Generating a Basic Sinusoid

Using MATLAB, a sinusoid was first generated with amplitude A = 1, frequency f = 1000 Hz, sampling frequency fs = 16000 Hz, duration = 0.01 s, and phase = 0. The discrete frequency is F = f/fs = 1000/16000 = 1/16 cycles/sample, so there are 1/F = 16 samples per cycle. The resulting plot showed a smoothly-varying 1 kHz sinusoid over 10 full cycles.

Changing the phase to 0.5 rad shifted the waveform horizontally in time, without changing its shape, amplitude, or frequency.

Repeating with a duration of 1s and the same phase, produced 1000 cycles over the plot width; as the manual predicted, the individual cycles could no longer be observed, and the plot appeared as a solid-colored block. However, the audible tone was a steady, pure 1 kHz note. Repeating again with the phase changed to 0.5 rad gave the same dense, solid-block plot, and no audible difference could be heard. A constant phase offset does not change the pitch of a steady tone.

Reducing the amplitude to 0.1 (duration = 1 s, phase = 0.5) produced the same dense plot, now confined to a smaller vertical range −0.1 to 0.1 instead of −1 to 1. The tone was noticeably quieter but at the same pitch, confirming that amplitude controls the loudness of the waveform, not its frequency.

2. Effect of Sampling Frequency

Part (a) was repeated with A = 1, phase = 0, duration = 0.01 s, at three lower sampling frequencies: 7000, 3000, and 1300 Hz. At fs = 7000 Hz, the waveform was visibly coarser (about 7 samples per cycle instead of 16) but still clearly recognizable as a 1 kHz sinusoid. At fs = 3000 Hz, with only about 3 samples per cycle, the plotted waveform began to visually distort and no longer resembled a smooth sine wave. At fs = 1300 Hz, below the 2000 Hz Nyquist rate required to represent a 1000 Hz signal, the waveform (Fig. 1) looked noticeably different from the original 1 kHz sinusoid, since there were no longer enough samples per cycle to capture the original frequency.

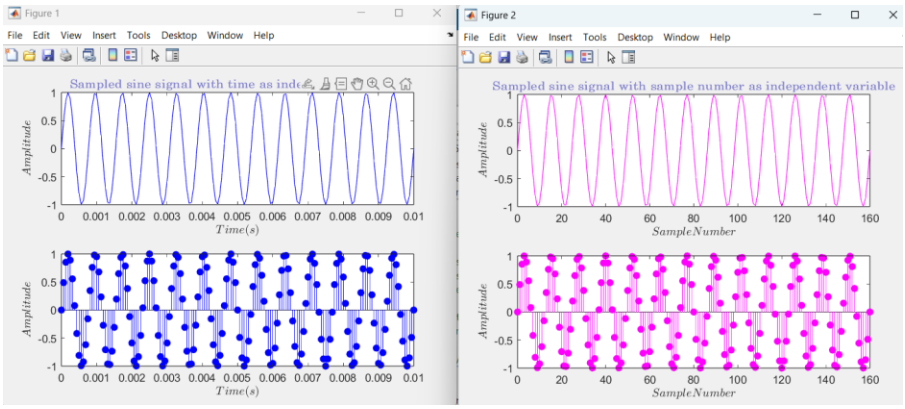


Figure 1: Sampled 1 kHz sinusoid with fs = 1300 Hz (below the Nyquist rate), showing visible aliasing distortion.

Repeating with 1-second durations at fs = 7000, 3000, 1700, 1300, and 1100 Hz, the tone did not sound the same across all of these rates. At the higher rates the tone still sounded close to the original 1 kHz pitch, but as fs dropped toward and below 2000 Hz (1700, 1300, 1100 Hz), the pitch changed noticeably from the original tone. This is explained by the Nyquist sampling theorem: once the sampling frequency falls below twice the signal frequency, the samples can no longer uniquely represent the original 1000 Hz tone, and what is actually heard is a lower aliased frequency that the same set of samples is equally consistent with.

3. Varying Amplitude and Phase

Three sinusoids with f = 1000 Hz, F = 1/16, and 80 samples (5 cycles) were generated with amplitudes A = 0.1, 0.3, and 1 and plotted together (Fig. 2). All three signals shared the same frequency and shape; only the peak-to-peak height scaled with A, confirming that amplitude controls the vertical scale (and, audibly, the loudness) of a sinusoid without affecting its frequency or shape.

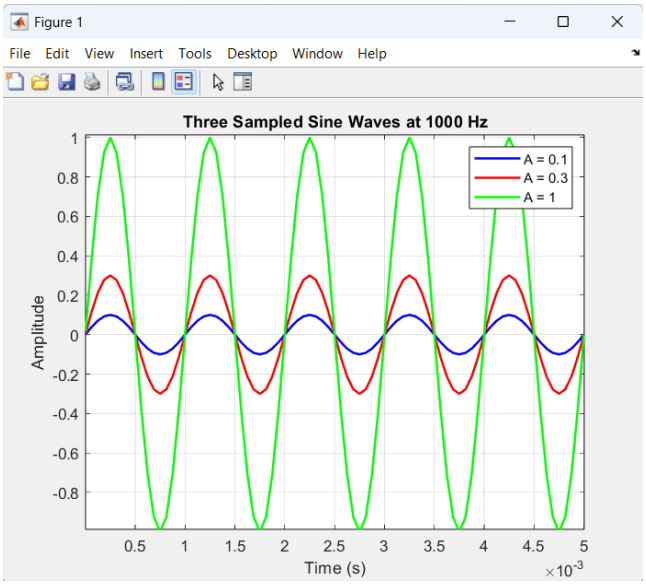


Figure 2: Three sinusoids at 1000 Hz with amplitudes A = 0.1, 0.3, and 1.

Next, three sinusoids with the same frequency and amplitude (A = 1) but phases θ = 0, π/4, and π/2 were plotted together (Fig. 3). Each waveform had the same shape and height, but was shifted horizontally relative to the others, confirming that phase controls the timing/starting point of a sinusoid rather than its magnitude or frequency.

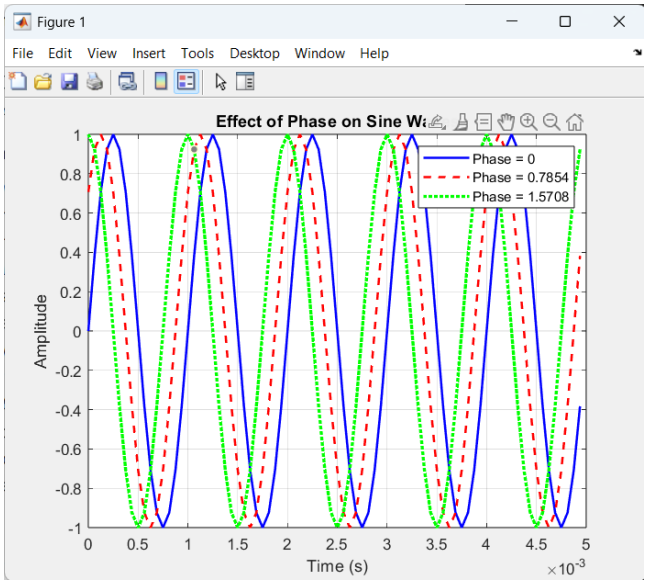


Figure 3: Three sinusoids at 1000 Hz with phases θ = 0, π/4, and π/2.

Repeating with θ = 0, π, and 4π showed that the θ = 0 and θ = 4π waveforms were identical, since phase is periodic with 2π (4π = two full cycles of phase), while θ = π produced a waveform that was the exact mirror image (inverted) of the θ = 0 case.

Finally, the same three phases (0, π/4, π/2) were generated at f = 1000 Hz with 8000 samples so they could be played as sound. No difference could be heard between the three. Phase does not affect the pitch of a steady sinusoid, only where in time it starts, which the ear cannot perceive for a continuous tone.

4. Combining Sinusoids

A sinusoid with amplitude $2/\pi \approx 0.637$, f = 1000 Hz, fs = 50000 Hz, and duration = 0.005 s was generated as a baseline sine wave. A second sinusoid, with amplitude $2/3\pi \approx 0.212$ at the third harmonic (3000 Hz), was then added to it. The sum began to show flattened peaks and steeper transitions compared to the original pure sinusoid.

Additional odd harmonics (5, 7, and 9 times the fundamental) were then added one at a time, each with amplitude $2/n\pi$. As each higher harmonic was included, the summed waveform's peaks and troughs flattened further and the transitions between them grew steeper, progressively taking on the flat-topped shape of a square wave. Fig. 4 shows the result after summing the fundamental plus the 3rd, 5th, 7th, and 9th harmonics (n = 1, 3, 5, 7, 9).

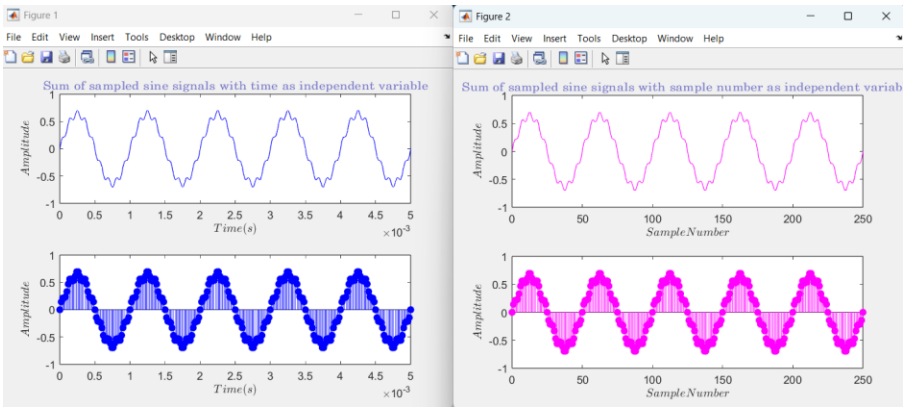


Figure 4: Sum of the fundamental (1 kHz) and its 3rd, 5th, 7th, and 9th harmonics, each weighted by 2/nπ, approximating a square wave.

Repeating with 1-second-long versions of each signal and listening to them: the pure 1 kHz sinusoid sounded like a clean, pure tone, while the summed signals sounded progressively brighter and harsher as more odd harmonics were added, approaching the characteristic buzzy sound of a square wave. This matches the time-domain results. The added high-frequency harmonics are what both sharpen the waveform's transitions and add the harsher quality to the sound.

5. Additional Exploration

As an additional, optional exercise, the odd harmonics from part 4 were summed again using a different weighting instead of 2/nπ (Fig. 5).

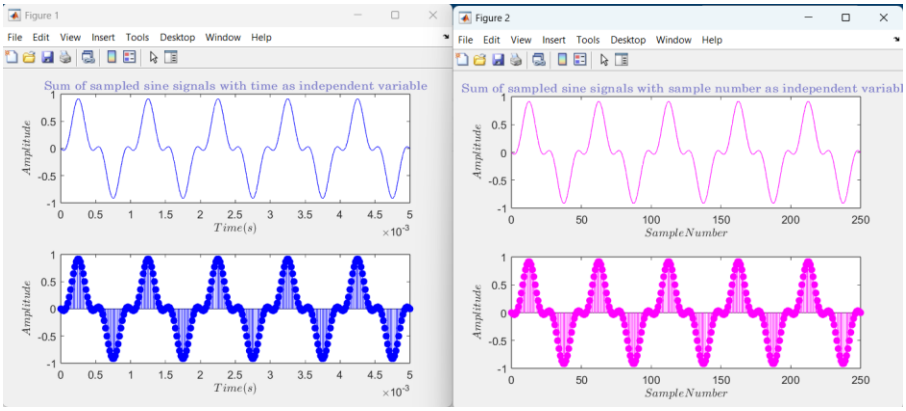


Figure 5: Sum of the same odd harmonics (1, 3, 5, 7, 9 kHz) using a different amplitude weighting than 2/nπ.

Each harmonic was given the same constant amplitude, 2/(3π). The resulting waveform did not flatten into the same square-wave shape as Fig. 4, but it kept a more rounded, multi-peaked profile within each half-cycle. This shows that it is specifically the decreasing 1/n amplitude law that produces the square-wave approximation. Weighting the harmonics equally keeps relatively more high-frequency content and does not flatten the peaks and troughs the way the proper square-wave weighting does.

Conceptually, keeping the same weighting but changing the relative phase between harmonics would not change which frequencies are present or how strong each one is, since phase differences only affect how the components align in time, not their amplitudes. Shifting the phases this way is expected to turn the flat-topped square wave into a lopsided, sawtooth-like shape, since the harmonics no longer reinforce each other at the same points in each cycle. Audibly, this version would be expected to sound close to the unshifted version, since human hearing is relatively insensitive to the phase relationships between harmonics in a steady tone.

III. Discussion and Conclusions

This lab showed, both visually and audibly, what each parameter of a sinusoid controls: amplitude sets the loudness and vertical scale of the waveform, phase shifts the waveform in time without changing its shape or sound, and frequency sets the pitch and how densely the waveform oscillates. Sampling frequency turned out to matter just as much as the signal's own frequency. Once fs dropped below twice the signal frequency, the samples no longer uniquely represented the original tone, and both the plotted waveform and the audible pitch changed to reflect an aliased frequency instead, a direct demonstration of the Nyquist sampling theorem.

Combining sinusoids showed the other side of the same idea. A square-like waveform could be built up from nothing but a 1 kHz fundamental and its odd harmonics, each scaled by $2/n\pi$, with the approximation looking more like a true square wave as more harmonics were added. This is the basic idea of Fourier synthesis that complex periodic signals can be expressed as, and reconstructed from, a sum of simple sinusoids at the right frequencies, amplitudes, and phases.

Overall, this lab connected the abstract parameters of a sinusoid (A, f, θ) to concrete, observable effects in both the time domain and by ear, thus demonstrating two ideas: aliasing and Fourier synthesis.

IV. References

[1] Dr. Iman Salama. “Lab 6 – Creating and Combining Sinusoids in MATLAB.” Northeastern University. 21 September 2023.